Derivative tests for local minimum/maximum

min/max $\Longleftrightarrow$ $f'(\theta) = 0$ (the backward arrow is struck through)

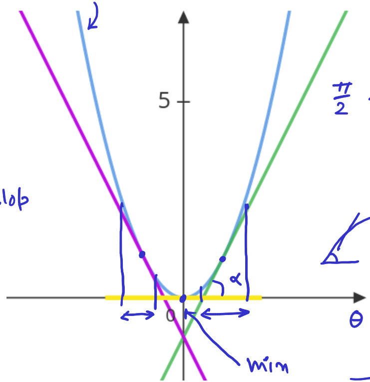
$\pi > \alpha > \frac{\pi}{2}$

$\alpha = 0$ or π

$\frac{\pi}{2} > \alpha > 0$

$f(\theta) = \theta^2$

Tan(α) = slop

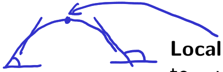


- Negative slop $f'(\theta) < 0$ means function is decreasing.

Zero slop $f'(\theta) = 0$ gives a critical point.

- Positive slop $f'(\theta) > 0$ means function is increasing.

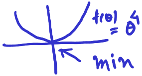
Local Minimum: Slop $f'(\theta)$ changes its sign from $-$ve to $+ve$, that is, $f''(\theta) > 0$ at a critical point.



Local Maximum: Slop $f'(\theta)$ changes its sign from $+$ve to $-ve$, that is, $f''(\theta) < 0$ at a critical point.

$f(\theta) = \theta^3$

If $f''(\theta) = 0$ at a critical point, then the test is inconclusive (there can be a **saddle point**, compare θ^3 and θ^4 at $\theta = 0$).



Convex function

$\underbrace{f''(\theta)}_{2} > 0$

Concave

$f''(\theta) < 0$

$f(\theta) = -\theta^2$